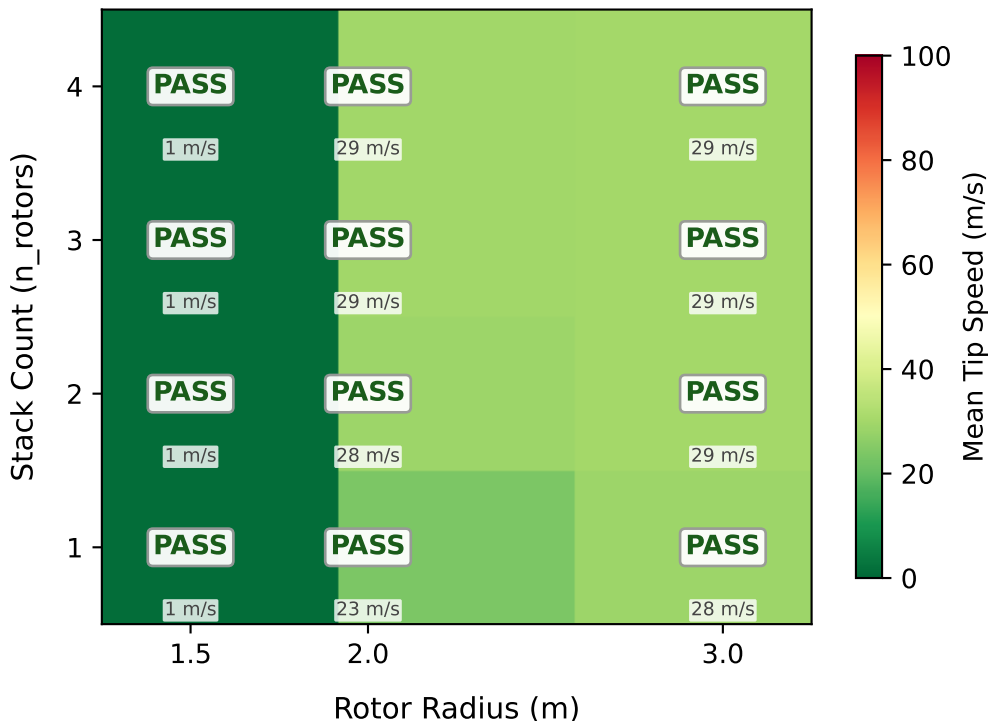


## Tip Speed — Noise Gate ( $\leq 100$ m/s)



*IMPLICATIONS: All configurations stay well below 100 m/s tip speed. Maximum tip speed across the sweep is  $\sim 39$  m/s ( $R=3.0$  m,  $N=4$  at 12 m/s). Tip speed saturates in the 20–39 m/s range for  $R \geq 2.0$  m.  $R=1.5$  m is stalled at 1–2 m/s. Noise is not a limiting factor. Data: BEM v2.1, corrected polygon solver. See SPEC.md §6.6.*